EVERY SMOOTH COVERING OF A SCHEME IS REFINED BY AN ETALE COVERING

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 10934-10945 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 10947-10954, the complete author proof (6 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The etale-neighbourhood-through-a-point lemma for smooth morphisms (lemma-etale-nbhd-dominates-smooth, same chapter, collected as L16018) supplying the local charts; cited input with the covering assembly written in the retained span.

Source reference. Original lemma 055U: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10846–10849. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 00VT: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10954–10954. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0215: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10954–10954. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 021Z: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10954–10954. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $\mathcal{U} = \{S_i \to S\}_{i \in I}$ be a smooth covering of S, see Topologies, Definition 021Z. Then there exists an étale covering $\mathcal{V} = \{T_j \to S\}_{j \in J}$ (see Topologies, Definition 0215) which refines (see Sites, Definition 00VT) $\mathcal{U}$.*

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. For every $s \in S$ there exists an $i \in I$ such that s is in the image of $S_i \to S$. By Lemma 055U we can find an étale morphism $g_s : T_s \to S$ such that $s \in g_s(T_s)$ and such that g_s factors through $S_i \to S$. Hence $\{T_s \to S\}$ is an étale covering of S that refines $\mathcal{U}$. □